How Our Project Works

A Simple, No-Jargon Explanation

A Multi-Protocol Cyber-Deception Framework with AI and Blockchain

---

The Big Idea in One Sentence

We build fake computer systems that look real to hackers. When a hacker attacks one, we secretly watch everything they do, use AI to reply convincingly so they keep going, and lock the evidence in a tamper-proof record nobody can secretly change.

> Real-life comparison — the fake shop. Imagine a shop owner tired of shoplifters. So they open a decoy shop full of fake goods, with hidden cameras everywhere. Thieves walk in thinking it's real, and every move they make is recorded — but nothing valuable is ever at risk. Our project is that decoy shop, but for computers.

Why Do We Need This?

Normal "traps" for hackers (called honeypots) have a weakness: they give the same fixed answers every time. A smart hacker quickly notices the answers are robotic and realises "this is a trap!" and leaves. It's like a mannequin in a shop — stand near it for a second and you know it isn't a real person.

We fix this with three upgrades:

1. Many fake traps	2. AI brain	3. Tamper-proof records
5 different fake systems, not 1	replies like a real system, so hackers stay fooled	evidence can't be secretly changed

Upgrade 1: Many Fake Traps (not just one)

Instead of one fake system, we run five — each pretending to be a different thing a company owns: a login server, a database, an AI service, a crypto wallet system, and a software-build server. This catches many kinds of attackers at once.

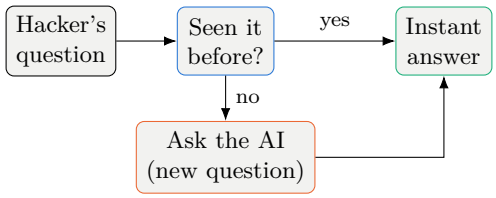
> Comparison — a house with decoy valuables in every room. Instead of one fake safe, you place a fake jewellery box in the bedroom, a fake wallet in the hall, a fake laptop in the study. Whichever room a burglar enters, a camera is already watching.

Upgrade 2: The AI Brain (the clever part)

When a hacker types something we've never seen before, older traps freeze or give a wrong answer — an instant giveaway. Instead, we ask an AI (a language model) to make up a realistic reply on the spot, exactly like a real computer would. The hacker never realises they're talking to a decoy.

Comparison — a fake receptionist who never breaks character. A cardboard cut-out receptionist fools no one. But a trained actor who answers any question naturally? You'd believe they're real. The AI is that actor.

To keep it fast and smart, we use a 3-step shortcut: common questions get instant pre-written answers; repeated questions are remembered; only brand-new questions go to the (slower) AI. Best of both worlds.



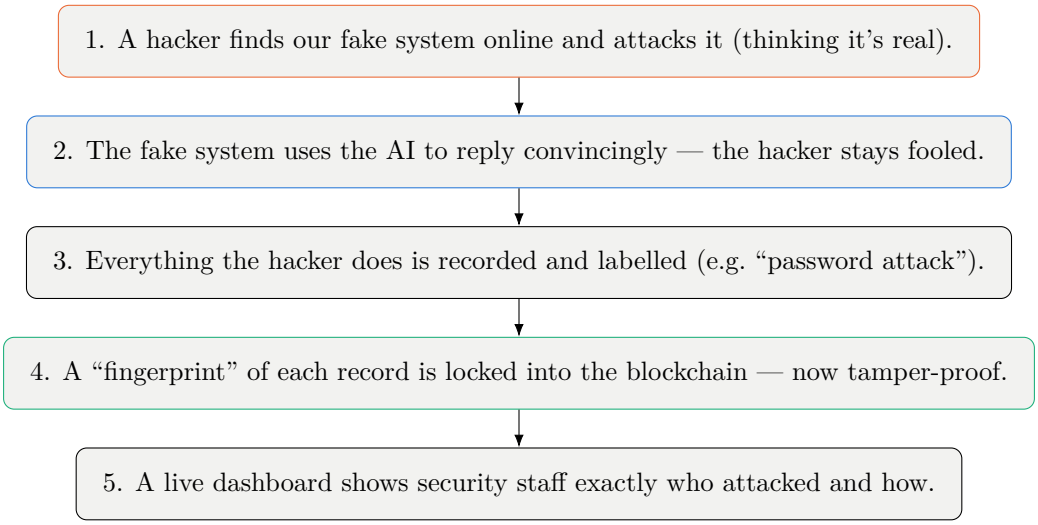
Upgrade 3: Tamper-Proof Records (blockchain)

Every action the hacker takes is recorded. But there's a catch: since the trap is being attacked, a clever hacker might try to delete or edit the record of what they did. So we take a unique "fingerprint" of each record and lock that fingerprint into a blockchain — a record that, once written, can never be secretly changed.

Comparison — a notary stamp / wax seal. Think of an important letter sealed with wax. If anyone opens or alters it, the seal breaks and everyone can see it was tampered with. Our system does this digitally: if even one letter of a record is changed later, the "fingerprint" no longer matches, and the tampering is caught instantly.

The Whole Story: One Attack, Start to Finish

Here is what happens, step by step, when a hacker attacks our system:



What Makes Ours Special

Most existing hacker-traps do only one of these things. Ours is the first to combine all three — many fake systems + a convincing AI brain + tamper-proof evidence — in a single package.

	Convincing (AI)	Many systems	Tamper-proof records
Old traps	No	No	No
Newer AI traps	Yes	Usually no	No
Our project	Yes	Yes	Yes

The One-Line Summary (for a viva)

> "We built five fake computer systems that use AI to convincingly fool attackers into revealing their methods, and we lock every piece of evidence into a blockchain so it can never be secretly tampered with — giving us honest, trustworthy intelligence about how real attacks happen."

Real terms (so you can connect the simple idea to the technical words):
Fake trap = honeypot | AI brain = large language model (LLM) | Fingerprint = SHA-256 hash | Tamper-proof record = blockchain